



The utility of machine learning algorithms for the prediction of patient-reported outcome measures following primary hip and knee total joint arthroplasty

Christian Klem¹ · Akachimere Cosmas Uzosike¹ · John G. Esposito¹ · Michael Joseph Harvey¹ · Ingwon Yeo¹ · Murad Subih¹ · Young-Min Kwon¹

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Abstract

Background Patient-reported outcome measures (PROMs) are increasingly used as quality benchmark in total hip and knee arthroplasty (THA; TKA) due to bundled payment systems that aim to provide a patient-centered, value-based treatment approach. However, there is a paucity of predictive tools for postoperative PROMs. Therefore, this study aimed to develop and validate machine learning models for the prediction of numerous patient-reported outcome measures following primary hip and knee total joint arthroplasty.

Methods A total of 4526 consecutive patients (2137 THA; 2389 TKA) who underwent primary hip and knee total joint arthroplasty and completed both pre- and postoperative PROM scores was evaluated in this study. The following PROM scores were included for analysis: HOOS-PS, KOOS-PS, Physical Function SF10A, PROMIS SF Physical and PROMIS SF Mental. Patient charts were manually reviewed to identify patient demographics and surgical variables associated with postoperative PROM scores. Four machine learning algorithms were developed to predict postoperative PROMs following hip and knee total joint arthroplasty. Model assessment was performed through discrimination, calibration and decision curve analysis.

Results The factors most significantly associated with the prediction of postoperative PROMs include preoperative PROM scores, Charlson Comorbidity Index, American Society of Anaesthesiology score, insurance status, age, length of hospital stay, body mass index and ethnicity. The four machine learning models all achieved excellent performance across discrimination (AUC > 0.83), calibration and decision curve analysis.

Conclusion This study developed machine learning models for the prediction of patient-reported outcome measures at 1-year following primary hip and knee total joint arthroplasty. The study findings show excellent performance on discrimination, calibration and decision curve analysis for all four machine learning models, highlighting the potential of these models in clinical practice to inform patients prior to surgery regarding their expectations of postoperative functional outcomes following primary hip and knee total joint arthroplasty.

Level of evidence Level III, case control retrospective analysis.

Keywords Hip and knee total joint arthroplasty · Patient-reported outcome measures · Machine learning · Artificial intelligence · Risk factors

Introduction

Total hip and knee arthroplasty (THA; TKA) represent one of the most commonly performed orthopaedic procedures to treat patients with end-stage osteoarthritis, and the utilization of total joint arthroplasty (TJA) procedures is forecasted to increase four-fold over the next decade [1]. Alongside the growing demand of TJA, there is an increased scrutiny regarding the value of delivered care. Over recent years,

✉ Young-Min Kwon
ymkwon@mgh.harvard.edu

¹ Bioengineering Laboratory, Department of Orthopaedic Surgery, Massachusetts General Hospital, Harvard Medical School, 55 Fruit Street, Boston, MA 02114, USA

there has been an increasing focus of public health officials to shift from volume-based, fee-for-service models to a patient-centered, value-based approach [2, 3]. The success of TJA has traditionally been measured from a surgeon's perspective utilizing outcome metrics such as implant survival and revision surgery [4]. However, the shift to value-based care has seen an increasing use of patient-reported outcome measures (PROMs), which quantify the clinical effectiveness of arthroplasty surgeries [5–8]. There are numerous PROM scores available in the literature to assess not only TJA performance, but also overall patient well-being and mental health [9–11]. The increasing use of PROMs has important administrative, clinical and financial implications, which highlight the clinical utility of predictive models to estimate postoperative PROM scores following hip and knee total joint arthroplasty.

Prior retrospective analyses investigated risk factors for poor patient-reported outcome measures following total hip and knee arthroplasty, demonstrating numerous patient and surgical factors to be associated with poor postoperative PROMs [12, 13]. However, these studies mainly focused on single risk factors and their association with post-operative PROM scores, thereby neglecting the complex relationship between patient and surgical factors. Machine learning (ML) methods have the capacity to effectively analyze non-linear relationships between risk factors in large and complex datasets, thereby achieving accuracies for risk factor predictions that outperform previously used statistical methods [14]. ML algorithms have been successfully introduced and adopted in many medical fields including cardiology, ophthalmology and dermatology [15]. Recent efforts in orthopaedics mainly explored ML models in terms of imaging-based diagnosis and the advancement of value-based care [16–18]. There is a limited number of studies that attempted to predict PROM scores following hip and knee total joint arthroplasty utilizing machine learning models [19–21]. However, these studies were limited by the consideration of only a single PROM score as well as a small sample size which led to PROM score predictions with only moderate accuracy. Therefore, this study aimed to develop and validate machine learning models for the prediction of numerous patient-reported outcome measures following hip and knee total joint arthroplasty.

Materials and methods

Patient cohort

The present analysis was conducted in concordance with the guidelines provided by the Transparent Reporting of a multivariable prediction model for Individual Prognosis or Diagnosis (TRIPOD). This study was approved by the

institutional review board. A retrospective chart review was performed to identify all patients who underwent primary total hip and knee arthroplasty at a single tertiary academic institution. All patients successfully completed postoperative patient-reported outcome measures at 1-year follow-up, including the HOOS-PS, KOOS-PS, Physical Function SF10A, PROMIS SF Physical and PROMIS SF Mental [22]. All patients also completed pre-operative PROM scores within 3 month prior to hip or knee total joint arthroplasty. Patients with simultaneous bilateral surgery, partial joint arthroplasty, emergency surgery and incomplete data were excluded from analysis in concordance with prior literature [19, 21]. A total of 4526 patients following total hip and knee arthroplasty were included for the development of machine learning models to predict postoperative patient-reported outcome measures.

Clinical variables

Patient and surgical factors associated with poor patient-reported outcome measures were collected using our institution's electronic medical record system [19]. These variables include (1) patient demographics (age, gender, body mass index [BMI], ethnicity, marital status, social status, insurance status); (2) medical comorbidities (American Society of Anesthesiologist Physical Status score [ASA score], Charlson comorbidity index [CCI]); (3) preoperative medications (opioid use); (4) surgical parameters (anesthesia type, tranexamic acid usage, component fixation method, operation time, blood loss, transfusion rates, tourniquet use).

Predictive machine learning models

We developed four state-of-the-art supervised machine learning methods: (1) artificial neural networks (ANN), (2) support vector machine (SVM), (3) random forest (RF), and (4) elastic-net penalized logistic regression (ENP). These four machine learning models were chosen for analysis based on previous literature demonstrating high accuracies for these candidate models in the prediction of clinical outcomes for patients following hip and knee total joint arthroplasty [23]. The primary THA and TKA datasets as shown in Table 1 were randomly divided into two datasets using an 80:20 split ratio to investigate the ability of these machine learning models to predict postoperative PROM scores [24]. These two datasets involve the training dataset (1709 THAs; 1911 TKAs) as well as the testing dataset (428 THAs; 478 TKAs). A five-fold cross-validation was performed to develop and assess all machine learning candidate models in the training set.

Metrics used to evaluate machine learning model performance included discrimination (area under the receiver operating characteristics curve [AUC]), calibration

Table 1 Patient cohort characteristics

Characteristic	Primary THA patients (<i>N</i> = 2137)	Primary TKA patients (<i>N</i> = 2389)
Demographics		
Age (years)	71.4 ± 16.8	72.2 ± 17.5
Gender	1043 males; 1094 females	1271 males; 1118 females
BMI (kg/m ²)	31.5 ± 3.9	32.0 ± 4.4
Laterality	1083 left; 1054 right	1228 left; 1161 right
ASA score (%)	ASA 1–93 (4.3%) ASA 2–1287 (60.2%) ASA 3–707 (33.1%) ASA 4–50 (2.3%)	ASA 1–89 (3.7%) ASA 2–1391 (58.2%) ASA 3–782 (32.7%) ASA 4–127 (5.3%)
Charlson comorbidity index	2.1 ± 1.6	1.9 ± 1.5
Insurance status (medicare; private)	1071; 1066	1161; 1228
Ethnicity (white, African, American, hispanic, Asian)	2065, 32, 27, 13	2288, 48, 36, 17
Follow-up time (years)	1.6 ± 1.0	1.5 ± 1.1
Comorbidities		
Smoking (%)	612 (28.6%)	754 (31.5%)
Drinking (%)	577 (27.0%)	780 (32.6%)
Drug abuse (%)	63 (2.5%)	82 (3.4%)
Diabetes mellitus (%)	225 (10.5%)	204 (8.5%)
Depression (%)	247 (11.5%)	191 (7.8%)
Renal failure (%)	167 (7.8%)	156 (6.5%)
Malignant tumor (%)	180 (8.4%)	229 (9.5%)
Hypertension	794 (37.1%)	818 (34.2%)
Surgical variables		
Blood loss (ml)	258 ± 132.6	104.6 ± 93.1
Operation time (min)	88.7 ± 18.5	95.7 ± 22.8
General anesthesia (%)	76.2	81.4
Tranexamic acid usage (%)	74.7	77.8
Blood transfusion rates (%)	4.7	7.1
Tourniquet use (%)	0	95.3
Cemented component fixation (%)	3.4	97.9

(calibration plot, intercept, slope), Brier score as well as decision curve analysis [25]. Relative variable importance plots were generated to identify the most important parameters for the machine learning model with the best overall performance. Interpretability of machine learning models was provided at both local and global levels [26].

Statistical analysis

All analysis was performed using Anaconda (Anaconda Inc., Austin, TX, USA), Matlab (MathWorks Inc., Natick, MA, USA), SPSS (SPSS Version 18.0, IBM Corp., Armonk, NY, USA), and Python (Python Software Foundation, Wilmington, DE, USA).

Results

Patient cohort

A total of 4526 patients (2137 THA; 2389 TKA) underwent primary total hip and knee arthroplasty. The patients had a mean age of 71.8 ± 17.1 years and a mean body mass index of 31.9 ± 3.2 kg/m². Patient demographics and surgical variables for both THA and TKA patient cohorts are summarized in Table 1.

Clinical variables

The machine learning models identified numerous clinical variables to be associated with the prediction of poor patient-reported outcome measures at 1-year following hip and knee total joint arthroplasty. These include preoperative PROM scores, Charlson Comorbidity Index, American Society of Anaesthesiology score (3 or 4), insurance status (Medicare), age (> 70 years), length of hospital stay (greater than 3 days), body mass index (> 35 kg/m²) and non-white ethnicity (Fig. 1). The strongest predictors for both THA and TKA cohorts were preoperative PROM scores, Charlson Comorbidity Index and American Society of Anaesthesiology score (Fig. 1).

Predictive machine learning model performance

In the training set of the THA patient cohort, the AUC for the candidate models ranged from 0.85 for support vector machines to 0.89 for neural networks (Table 2). The best calibration intercept was achieved by neural networks (intercept of 0.07; Table 2; Fig. 2). The lowest Brier score error was achieved by neural networks (Brier score of 0.051) and the highest Brier score error was obtained by support vector machines (Brier score of 0.055; Table 2). In the testing set of THA patients, the AUC for the four candidate models ranged from 0.84 to 0.87 (Table 3). The highest AUC was achieved by neural networks (AUC=0.87; Table 3). The Brier score errors in the testing set varied between 0.052 and 0.055. The lowest Brier score error was obtained by neural networks (Brier score of 0.052, Table 3). The machine

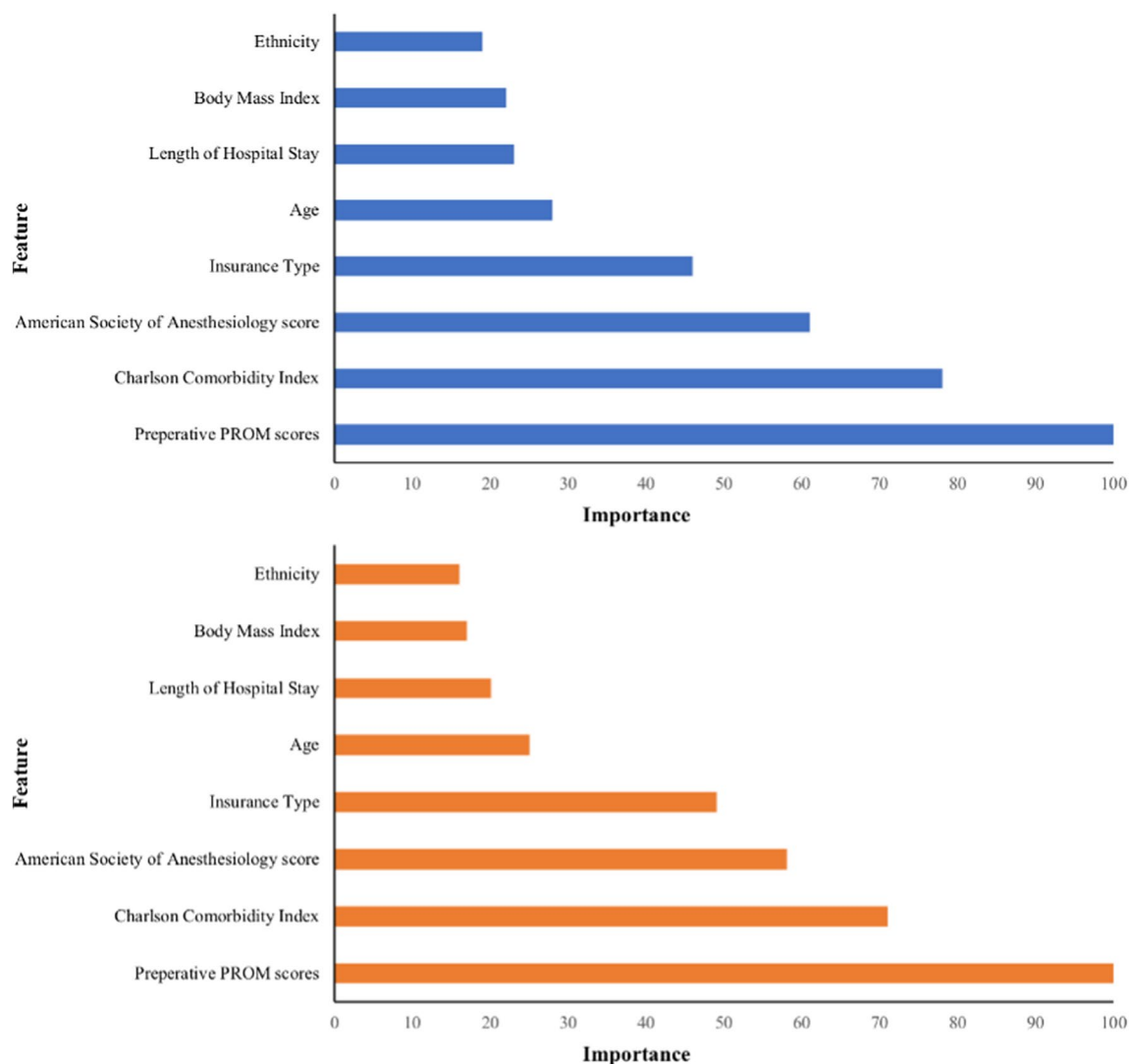


Fig. 1 Global variable importance plot for the prediction of patient-reported outcome measures in patients following hip (top) and knee (bottom) total joint arthroplasty

Table 2 Discrimination and calibration of machine learning algorithms on training set for THA patients

Metric	Neural network	Support vector machine	Random forest	Elastic-net penalized logistic regression
AUC	0.89 (0.87 to – 0.91)	0.86 (0.84 to – 0.88)	0.85 (0.82 to – 0.89)	0.87 (0.85 to – 0.90)
Intercept	0.07 (–0.02 to 0.17)	0.16 (0.04 to 0.28)	0.14 (0.07 to 0.21)	0.10 (– 0.01 to 0.21)
Slope	1.07 (0.99 to 1.15)	0.89 (0.75 to 1.03)	1.17 (0.98 to 1.36)	1.12 (1.02 to 1.22)
Brier	0.051 (0.049 to 0.054)	0.055 (0.053 to 0.057)	0.054 (0.051 to 0.058)	0.053 (0.052 to 0.054)

Data expressed as mean (95% confidence interval). Null model Brier score=0.057

learning models provided a prediction for PROM scores that was within 1.5 points of the actual PROM scores reported by the THA patients in 96.6% of the cases.

In the training set of TKA patients, the machine learning models demonstrated excellent discriminatory abilities. The AUC for the candidate models ranged from 0.84 for support vector machines to 0.88 for neural networks (Table 4). The lowest Brier score error was achieved by neural networks (Brier score of 0.052). In the testing phase of TKA patients, the AUC for the four candidate models ranged from 0.83 to 0.87 (Table 5). The highest AUC was achieved by neural networks (AUC=0.87; Table 5). The Brier score errors in the testing set varied between 0.053 and 0.058, with the lowest Brier score error for neural networks (Brier score of 0.053, Table 5). The machine learning models provided a prediction for PROM scores that was within 1.5 points of the actual PROM scores reported by the TKA patients in 96.1% of the cases.

In terms of model performance, there was no significant difference for machine learning candidate models predictions between THA and TKA patients considering the AUC ($p=0.56$), calibration intercept ($p=0.41$), calibration slope ($p=0.17$) and Brier score ($p=0.33$). The decision curve analysis demonstrated that the four candidate models all achieved higher net benefits for the prediction of postoperative patient-reported outcome measures for both THA and TKA patients, when compared to the default strategies of changing management for all patients or no patients (Fig. 3).

Discussion

This study developed and validated machine learning models for the prediction of postoperative patient-reported outcome measures for patients following primary total hip and knee arthroplasty. The study findings demonstrate excellent model performance for all four machine learning models. The strongest clinical parameters for the prediction of postoperative PROM scores included preoperative PROM

scores, Charlson Comorbidity Index and American Society of Anaesthesiology score. As these healthcare reforms continue to evolve, quantifying the value of surgical procedures is increasingly important. An important tool to assess the value of surgical interventions is represented by patient-reported outcome measures (PROMs), with most value-based bundled payment models now requiring the collection of pre- and postoperative PROM scores [27]. Through direct assessment of patients' perceptions of their health conditions, PROM scores capture an aspect of care that is most relevant to patients, thereby potentially providing important feedback on the effectiveness of a surgical intervention, while also aiding to improve the quality of care. Consequently, numerous PROMs were utilized in the recent literature to assess not only implant performance, but also overall patient well-being, physical performance levels as well as mental health in patients following hip and knee total joint arthroplasty [5, 6, 28].

There has been an increasing use of cost-effective computational tools, such as machine learning models, in the healthcare sector due to the decreasing costs of computational power [29]. Recent review articles highlighted the excellent accuracy of machine learning models for the prediction of patient outcomes across a variety of applications [29]. Machine learning algorithms have been successfully introduced and adopted in many medical fields [15, 30, 31]; however, this technology is relatively new to the field of orthopaedic surgery. There is great future potential as machine learning algorithms allow for a prediction of post-operative complications and outcomes on a patient-by-patient basis within seconds. Nonetheless, there is a paucity of studies utilizing machine learning models to predict numerous patient-reported outcome measures for patients following total hip and knee arthroplasty. Published literature to date focussed on the use of a single PROM score [19–21], thereby not considering the variety of PROM scores available to measure the clinical effectiveness of arthroplasty surgery. Additionally, these prior machine learning studies all failed to achieve excellent performance for model

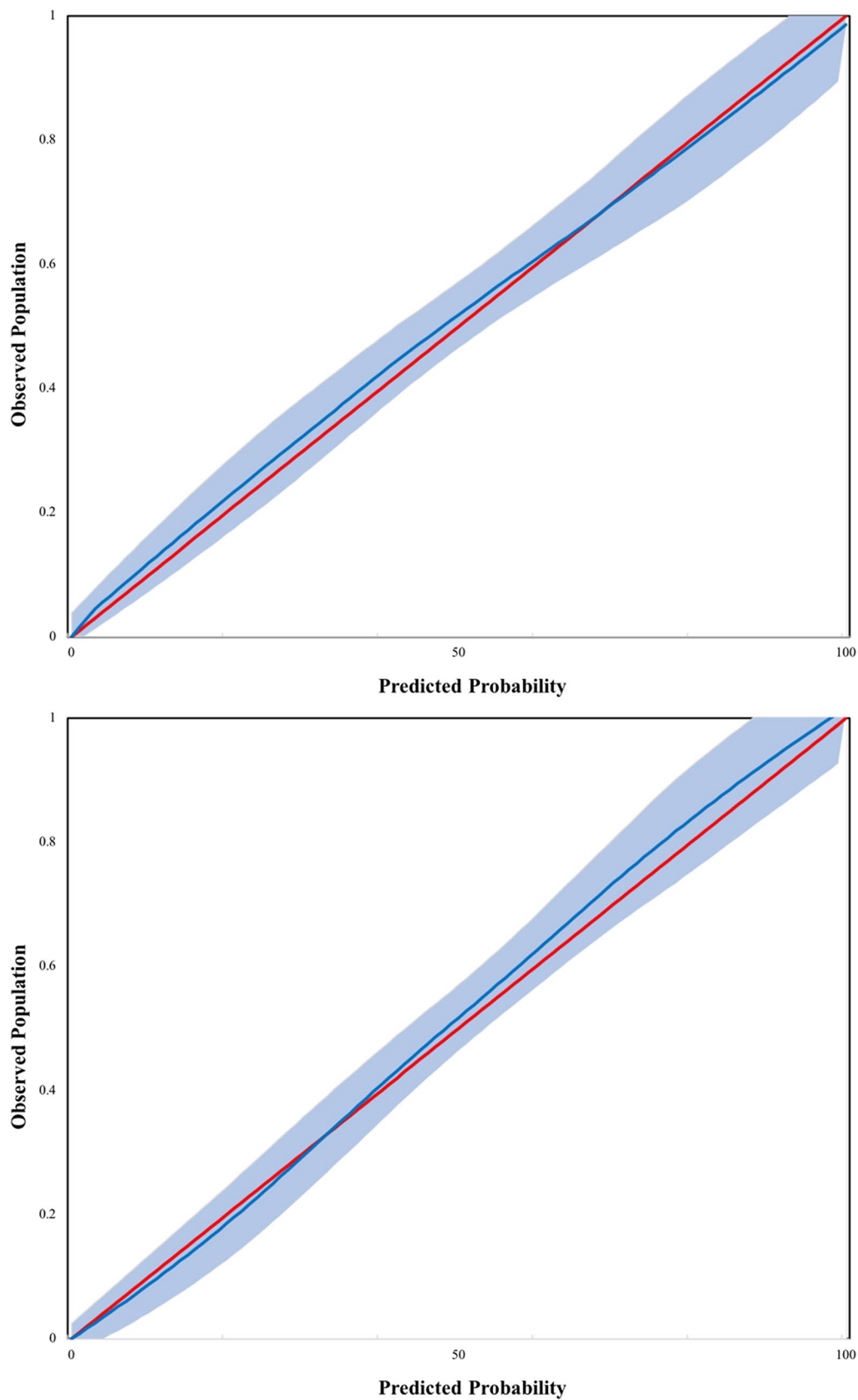


Fig. 2 Calibration plot for the neural network model for the prediction of patient-reported outcome measures in patients following hip (top) and knee (bottom) total joint arthroplasty

Table 3 Discrimination and calibration of machine learning algorithms on testing set for THA patients

Metric	Neural network	Support vector machine	Random forest	Elastic-net penalized logistic regression
AUC	0.87	0.84	0.85	0.86
Intercept	0.09	0.20	0.17	0.15
Slope	1.11	0.87	1.21	1.16
Brier	0.052	0.055	0.055	0.054

Data expressed as mean (95% confidence interval). Null model Brier score=0.058

Table 4 Discrimination and calibration of machine learning algorithms on training set for TKA patients

Metric	Neural network	Support vector machine	Random forest	Elastic-net penalized logistic regression
AUC	0.88 (0.86 to – 0.90)	0.84 (0.82 to – 0.86)	0.84 (0.83 to – 0.86)	0.87 (0.86 to – 0.89)
Intercept	0.04 (– 0.01 to 0.09)	0.22 (0.07 to 0.37)	0.16 (0.08 to 0.24)	0.08 (0.02 to 0.14)
Slope	1.09 (1.03 to 1.16)	0.90 (0.78 to 1.05)	1.15 (0.96 to 1.34)	1.15 (1.01 to 1.29)
Brier	0.052 (0.050 to 0.055)	0.056 (0.053 to 0.058)	0.055 (0.053 to 0.057)	0.053 (0.052 to 0.055)

Data expressed as mean (95% confidence interval). Null model Brier score=0.057

Table 5 Discrimination and calibration of machine learning algorithms on testing set for TKA patients

Metric	Neural network	Support vector machine	Random forest	Elastic-net penalized logistic regression
AUC	0.87	0.83	0.83	0.85
Intercept	0.08	0.25	0.14	0.16
Slope	1.10	0.84	1.19	1.16
Brier	0.053	0.058	0.056	0.055

Data expressed as mean (95% confidence interval). Null model Brier score=0.059

predictions [19–21]. Huber et al. developed a supervised machine learning model to predict the Oxford Hip and Knee score in 13,945 patients following primary hip and knee total joint arthroplasty, reporting an AUC of 0.73 for the predictive model [21]. In a different study, Sanchez-Santos et al. validated a machine learning model for the prediction of postoperative Oxford knee scores reporting an AUC of 0.69. The study by Sanchez-Santos et al. included a sample size of 1649 patients following total knee arthroplasty [19].

The present study identified numerous patient factors to be associated with the prediction of PROM scores at 1-year following total hip and knee arthroplasty. These include preoperative PROM scores, Charlson Comorbidity Index, American Society of Anaesthesiology score, insurance status, age, length of hospital stay, body mass index and ethnicity. These observations are in concordance with prior retrospective studies investigating risk factors for poor functional outcomes following total joint arthroplasty [32]. In a retrospective study including 562 unilateral patients

following TKA, Berliner et al. reported that the preoperative PROM scores were the strongest predictors for postoperative function as quantified through the Knee Injury and Osteoarthritis Outcome Score [13]. This highlights the strong influence of preoperative functional ability on postoperative functional outcomes, which has also been supported by numerous other studies in the literature [33–35]. The effect of Charlson Comorbidity Index and American Society of Anaesthesiology score on postoperative functional outcomes was illustrated by Cher et al., reporting a negative correlation between both metrics and postoperative PROM scores (Oxford Knee Score; SF-36 Physical Function) at 2-year follow-up, in a study cohort with 7532 patients following TKA [36]. Similar observations were reported in numerous other studies in the literature, supporting older age, obesity, Medicare insurance and non-white ethnicity as risk factors for poorer PROM scores following index TKA [37–40].

Although there is concurrence between the present study and prior retrospective studies in terms of risk factors for

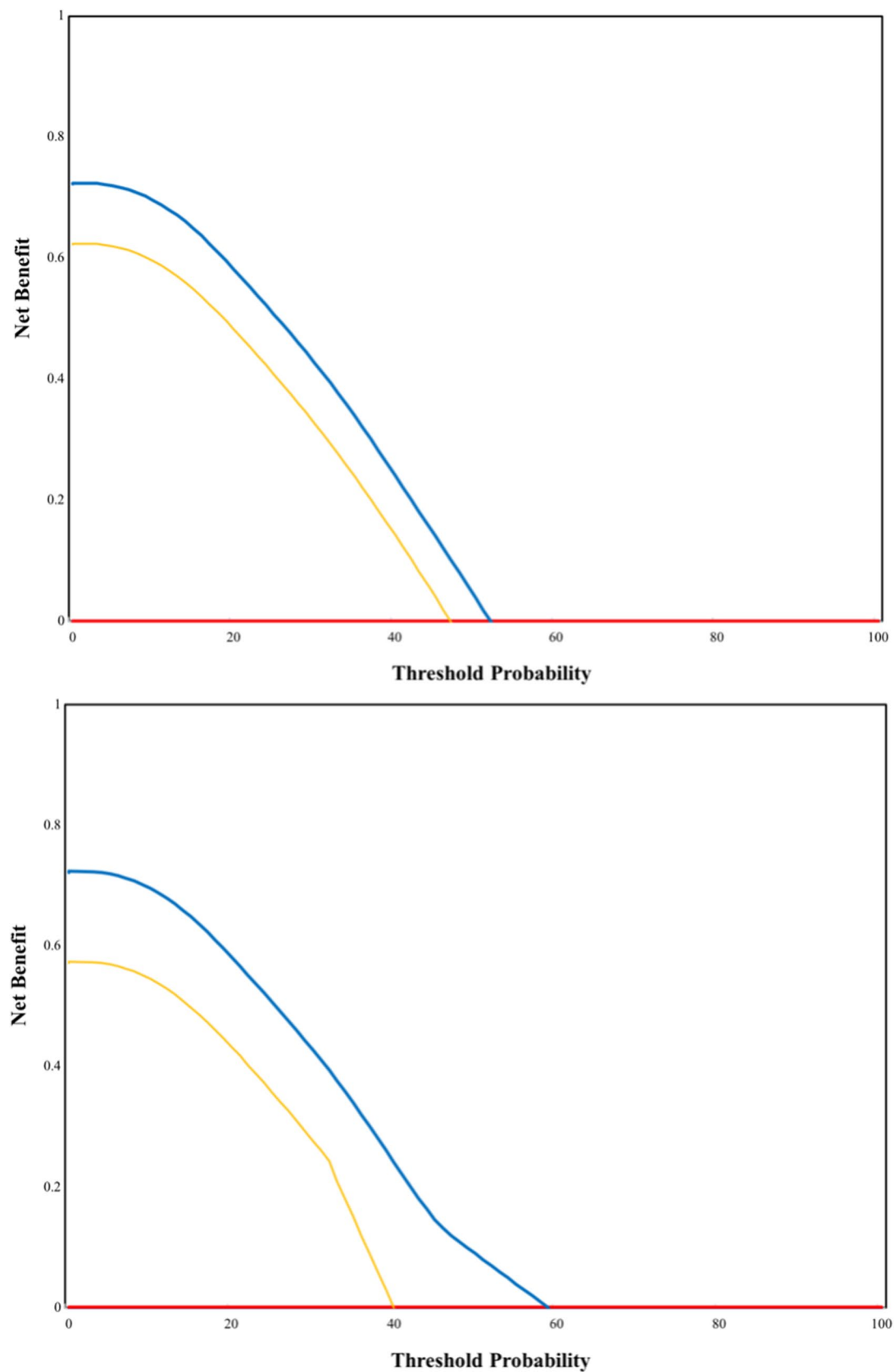


Fig. 3 Decision curve analysis for THA patients (top) and TKA patients (bottom) using neural network models showing the net benefit of the neural network model (yellow) relative to the default strategies of changing management for all patients (blue) or for no patients (red)

poor postoperative PROM scores [36, 41], the present machine learning analysis demonstrated that Charlson Comorbidity Index and the American Society of Anaesthesiology score were both amongst the three strongest predictors for poor postoperative PROMs. This highlights a stronger importance of these two patient factors than previously identified. Cher et al. reported an only moderately strong negative correlation between both metrics and postoperative PROM scores (Oxford Knee Score; SF-36 Physical Function) at a follow-up of 2 years post primary TKA [36]. Similarly, Elmallah et al. reported an only moderately strong effect of Charlson Comorbidity Index and American Society of Anaesthesiology score on postoperative PROM scores, with a limited number of comorbidities (cardiovascular disease, endocrine disease, gastrointestinal disease) representing the strongest risk factors [41]. The increased weight of Charlson Comorbidity Index and American Society of Anaesthesiology score as presented in this current study may be due to the use of machine learning models, which have been reported to yield higher accuracies for data analysis [29, 42].

The clinical utility of the validated machine learning models as presented in this study is based on a rigorous model evaluation as well as the high accuracy of the machine learning models as demonstrated during model validation. In the present study, the machine learning models provided a prediction for PROM scores that were within 1.5 points of the actual PROM scores reported by the THA/TKA patients in 96.6/96.1% of the cases. This is superior to prior studies, failing to achieve similar accuracies in 90% of arthroplasty patients [19–21]. Besides the high accuracy of the machine learning models of the present study, the performance of the four machine learning models on internal validation was verified through a rigorous evaluation in concordance with the Transparent Reporting of multivariable prediction models for Individual Prognosis or Diagnosis (TRIPOD) [23]. The clinical utility of machine learning models is based on its ability to predict PROM scores, both pre-operatively as well as post-operatively. This facilitates the identification of patients that may not or may only achieve limited benefit from TJA, as well as those patients that are likely to achieve a significant improvement from pre- to post-operative PROM scores. As patient satisfaction is not only associated with high PROM scores but rather with the absolute improvement in PROM scores from pre-to post surgery, identifying the potential of patients to improve PROMs has the ability to assist clinicians in pre-operative patient counselling.

The findings of this study have to be interpreted in light of its limitations. Firstly, this study represents a retrospective study including THA and TKA patients from a single large tertiary referral center, thus the generalizability of these machine learning models in clinical practice may be limited. Secondly, most of the patient factors that were

considered for analysis were binary. Therefore, this present study does not evaluate the effect of disease severity postoperative PROM scores. However, similar limitations were reported in prior retrospective studies on this topic [19, 20]. Thirdly, PROM scores have the potential for voluntary response or nonresponse bias as well as the inherent limitation of not capturing the importance a patient may place on a given outcome and its impact on their life. However, all PROM scores utilized in this presented study have been clinically validated [5, 43]. Finally, this study did not apply the machine learning models to identify patients which achieved minimally clinically important difference following TKA and THA [20]. However, this can easily be extracted from the model predicted postoperative PROM scores.

In conclusion, the current study developed and validated machine learning models for the prediction of postoperative patient-reported outcome measures following primary total hip and knee arthroplasty. The study findings illustrate excellent performance for all machine learning models, highlighting the potential of these machine learning models to inform patients prior to surgery regarding their expectations of postoperative functional outcomes following hip and knee total joint arthroplasty.

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

Informed consent Informed consent was obtained for the retrospective patient chart review.

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